

The oldest surviving sundial found in Egypt around 1500 BC

The earliest surviving sundial is located in the Egyptian museum in Berlin, with the catalog number 19744. Probably, it was found in Eschmunein. It dates from the reign of **Thutmosis III** (1479 - 1425 BC) during the New Kingdom. The name and titles of the pharaoh are engraved on the side

It is made of stone (schist) and is shaped like the letter L. The short piece is the gnomon (vertical rod giving a shadow). In the gnomon there is a hole for attaching **a small plumb bob**. Below is a vertical groove to align the thread of the plumb bob.

In the horizontal plane of the long bar there are five circles as marks of a time scale. The distances between the marks are proportional to **the numbers 1,2,3,4,5**.

The hypothesis of Ludwig Borchardt and Evert Bruins

In 1910 the German Egyptologist Ludwig Borchardt formulated the hypothesis that the shadow of the gnomon gives an indication of the so-called **unequal hours** on the time scale.

Those divide **the period between sunrise and sunset into twelve equal parts**. Because in wintertime the period between sunrise and sunset is shorter than in summertime, the hours, so determined, are unequal in length depending on the season. Unequal hours were in use until the Middle Ages.

The height of the sun varies also with the seasons and hence the length of the shadow. To give a shadow to the correct mark in every season, the gnomon should be variable in height. Therefore Borchardt assumed



Photo: Website Verein zur Förderung des Ägyptischen Museums Berlin

